Gerald Truong

Senior Software Engineer |
AI-Enabled Applications |
Full-Stack & Distributed Systems

Austin, TX
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Profiles

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https://www.researchgate.net/profile/Gerald-Truong

Skills

AI & Intelligent Automation
Advanced
Agentic AI Workflows, AI-Assisted Data Mapping, LLM Integration, Automation Solutions

Backend Development & APIs
Advanced
TypeScript, Node.js, NestJS, REST APIs, GraphQL, SSO, SQL, Ruby on Rails, PHP, Python

Frontend & UI/UX Engineering
Advanced
JavaScript (ES6+), React.js, Redux, Vue.js, Angular, HTML5, CSS3, Storybook, Design Systems

Data Engineering & Visualization
Advanced
D3.js, Visualize.js, Big Data UI Components, ETL Pipelines, SVG, Data Pipeline Monitoring

EdTech & Standards
Advanced
SIS/LMS Integrations, LTI 1.1 / 1.3, Rostering APIs, Canvas, Moodle, Brightspace, Blackboard

DevOps, Testing & Tools
Intermediate
Git, Docker, Puppeteer JS (E2E Testing), Webpack, Salesforce Platform (Apex), Pendo Analytics, JIRA

Hardware & Systems Engineering
Advanced
Signal Integrity, High-Speed Digital Design, Circuit Analysis, IBIS Modeling, CAD Application Development, FPGA, Verilog

Senior Software Engineer with extensive experience designing and delivering scalable enterprise software, APIs, data-intensive systems, and modern web applications across healthcare, education technology, cybersecurity, and business intelligence. Recently focused on building AI-enabled applications that leverage agentic workflows and intelligent automation to solve complex enterprise challenges. Progressive adopter of AI-assisted coding practices, optimizing development output through strategic application of agent technologies and intelligent automation tools. Proven track record of leading technical initiatives, architecting production-ready software, mentoring engineers, and delivering customer-focused solutions. Backed by a Master's degree in Electrical Engineering and deep expertise in high-speed digital design and circuit analysis, bringing a systems-thinking approach to designing reliable, scalable software and AI solutions. Fluent in both French and English.

Experience

Edlink
June 2023 to November 2025
Software Engineer / Technical Account Solution Engineer — Austin, TX
https://ed.link

- Architected and deployed an AI-assisted data mapping agent in Python and FastAPI, using the OpenAI SDK and agentic workflows to map clients' custom SIS data to Edlink's standard schema.
- Built a RAG-based support chatbot using the OpenAI SDK and pgvector for embedding storage and semantic retrieval, integrating LLM services into customer-facing operational workflows.
- Built GitHub Actions CI/CD pipelines for automated deployment and end-to-end testing of AI services and integrations.
- Engineered rostering, SSO, and coursework API integrations for Facts SIS, Blackbaud SIS, Toddle LMS, and Moodle LMS using Node.js, NestJS, and TypeScript.
- Developed LTI 1.1 and LTI 1.3 solutions across major LMS platforms including Canvas, Schoology, Brightspace, and Blackboard.
- Built an audit monitoring system with email notifications, plus data pipeline monitoring UIs and custom JSON property editors for rostering records in Vue.js.
- Triaged and resolved critical (P1) integration escalations by isolating root cause across Edlink, client applications, and third-party LMS/SIS providers, diagnosing logs and owning client communication through resolution as the formal engineering escalation point for Client Success.
- Monitored integration health and data quality across client rosters, investigating sync failures and missing users, classes, and enrollments to identify recurring provider-specific issues before they escalated.
- Served as the embedded technical authority for Client Success, bridging CSMs, clients, and Edlink Engineering/Product on all integration-related technical matters.
- Guided EdTech clients through implementation planning and their first five school/district onboardings, defining required integrations (SSO, rostering, grades, coursework) and coordinating milestones across client and Edlink engineering teams.

Medidata
2017 to March 2023
Senior Software Engineer — Austin, TX
https://www.medidata.com/en/

- Led a team of two data scientists to develop an OCR and ETL-driven budget forecasting data pipeline for clinical trial management.
- Architected core components for 1XDS, Medidata's enterprise React design system and Storybook component library.
- Designed and built template-based, quick-entry budgeting applications and complex clinical trial user workflows using React and Redux.
- Pioneered end-to-end testing frameworks across multiple core applications using Puppeteer JS.

- Refactored two Angular 1.5 applications to React JS using the Strangler Fig pattern
- Mentored junior developers in modern React/Redux practices.
- Integrated Pendo Analytics live chat features to enhance customer support.

Synack **2015 to 2017**
Senior Software Engineer Austin, TX
https://synack.com

- Engineered high-performance data visualization components for Synack's security dashboard product using React, D3.js, SVG, and REST APIs.
- Developed the core Synack platform messaging application utilizing React, Redux, and Ruby on Rails.
- Translated complex security dashboard design mockups into fully responsive, production-ready web applications.

Jobscience **2013 to 2015**
Technical Lead Austin, TX
https://www.linkedin.com/company/jobscience

- Developed mobile job board applications, candidate resume evaluator tools, and custom PDF viewing renderers on the Salesforce platform using Apex, JavaScript, HTML, and CSS.
- Provided technical leadership and code reviews for an offshore engineering team based in Belarus.

Jaspersoft **2011 to 2013**
Senior Software Engineer Austin, TX
https://www.jaspersoft.com

- Created and developed Visualize.js, an award-winning JavaScript framework for embedding interactive dashboards and reports into external web applications.
- Developed big data UI table components capable of rendering over 100,000 tabular data rows efficiently in the browser.
- Engineered cross-browser charting components and reduced overall CSS payload sizes by 25% across Jaspersoft Server web apps.

Isola **2010 to 2011**
Software Contractor (contract) Austin, TX
https://www.isola-group.com/

- Architected IsoDesign, the industry's first Web CAD application for high-speed PCB stackup design using JavaScript, SVG, Raphael.js, and MVC design principles.

Force 10 Networks **2006 to 2007**
Signal Integrity Engineer Austin, TX
https://en.wikipedia.org/wiki/Force10

- Executed signal integrity simulations and circuit analysis for 24-layer linecards featuring multi-gigahertz ASICs, DDR2+, QDR2+, and high-speed SerDes channels.
- Procured, validated, and maintained complex IBIS models for high-speed digital board design verification.

Northrop Grumman **2003 to 2005**
Electrical Engineer / Application Support Engineer Austin, TX
https://www.northropgrumman.com/

- Conducted circuit analysis and signal integrity simulations on critical digital modules for the F-22 software radio system and satellite C&DH; subsystems.
- Created and led training programs for Mentor Graphics signal integrity simulation software serving 40+ hardware engineers.

Motorola Computer Group **2000 to 2002**
Signal Integrity Engineer Tempe, AZ
https://emuseum.tempe.gov/objects/2767/motorola--2900-s-diablo-way--tempe-arizona

- Performed signal integrity analysis and IBIS model validations using Mentor Graphics and Cadence SI suites to guarantee signal quality on complex PCB layouts.
- Collaborated directly with IC vendors to acquire and review high-speed component simulation models.

Education

The University of Texas at Austin (McCombs School of Business) — **January 2024**
Austin, TX
Post Graduate Program in AI and Machine Learning

California State University, Long Beach — Long Beach, CA
Master's in Electrical Engineering